Software Requirements Specification for

Online Examination and Evaluation System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the requirements for the **Online Examination & Evaluation System**. The purpose of this document is to clearly define the system's functional and non-functional requirements so that they can be understood by project supervisors, developers, testers, and other stakeholders.

The requirements specified in this document are derived from structured analysis and design models, including **Use Case Diagram, Context Diagram, Data Flow Diagrams (Level-0 and Level-1), Activity Diagrams, Sequence Diagrams, and Class Diagram**. This document serves as a baseline agreement between stakeholders and developers regarding system functionality.

1.2 Document Conventions

This document follows standard Software Engineering and IEEE SRS conventions:

- UML diagrams are used to represent system behavior and structure.
- Data Flow Diagrams (DFD) are used to model data movement within the system.
- Diagram notations comply with UML 2.x and DFD standards.
- All diagrams are created using **Visual Paradigm**.
- Functional requirements are described using clear and precise language.

1.3 Intended Audience

This document is intended for:

- Faculty members and project evaluators
- Software developers implementing the system
- Testers responsible for system verification
- Students working on system enhancement

1.4 Product Scope

The **Online Examination & Evaluation System** is a web-based application designed to automate and manage the complete lifecycle of online examinations, starting from exam creation to evaluation and result generation. The system aims to provide a reliable, efficient, and secure platform for conducting examinations while minimizing manual intervention and administrative overhead.

The system supports three primary user roles: **Student, Teacher, and Admin**, each having distinct responsibilities and access privileges. Students can participate in online examinations and view results, teachers can create and evaluate examinations, and administrators can manage users, exams, and generate reports.

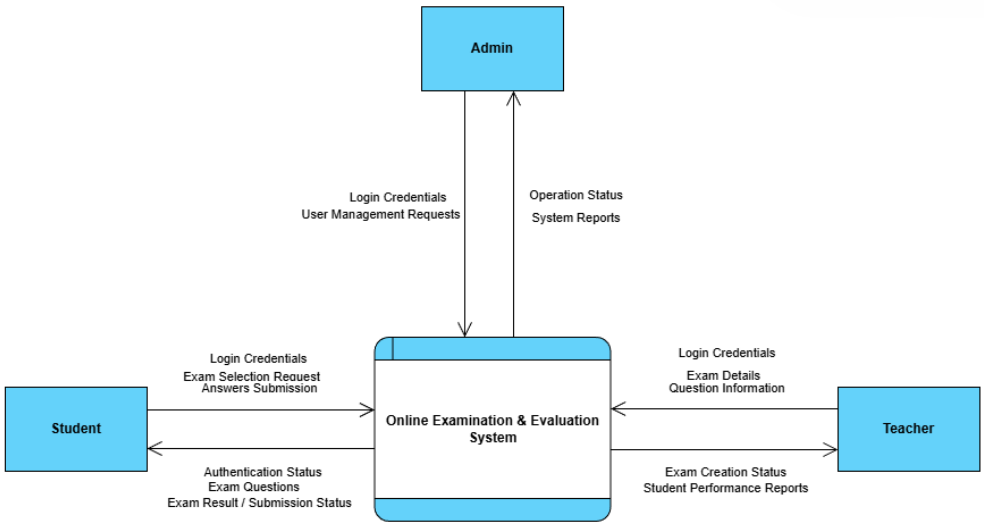
The overall scope and system boundary are represented using the **Context Diagram** and **Use Case Diagram**.

Context Diagram:

The Context Diagram defines the system boundary and shows how external entities interact with the system at a high level.

Field	Description
Actors	Student, Teacher, Admin
Description	The context diagram defines the system boundary and shows the interaction between the Online Examination & Evaluation System and external entities through data flows.
Data	Login credentials, exam details, questions, answers, results, reports

Stimulus	External entities send requests or data to the system
Response	System processes the data and returns responses such as results or confirmations
Comments	The system is represented as a single process with no internal details

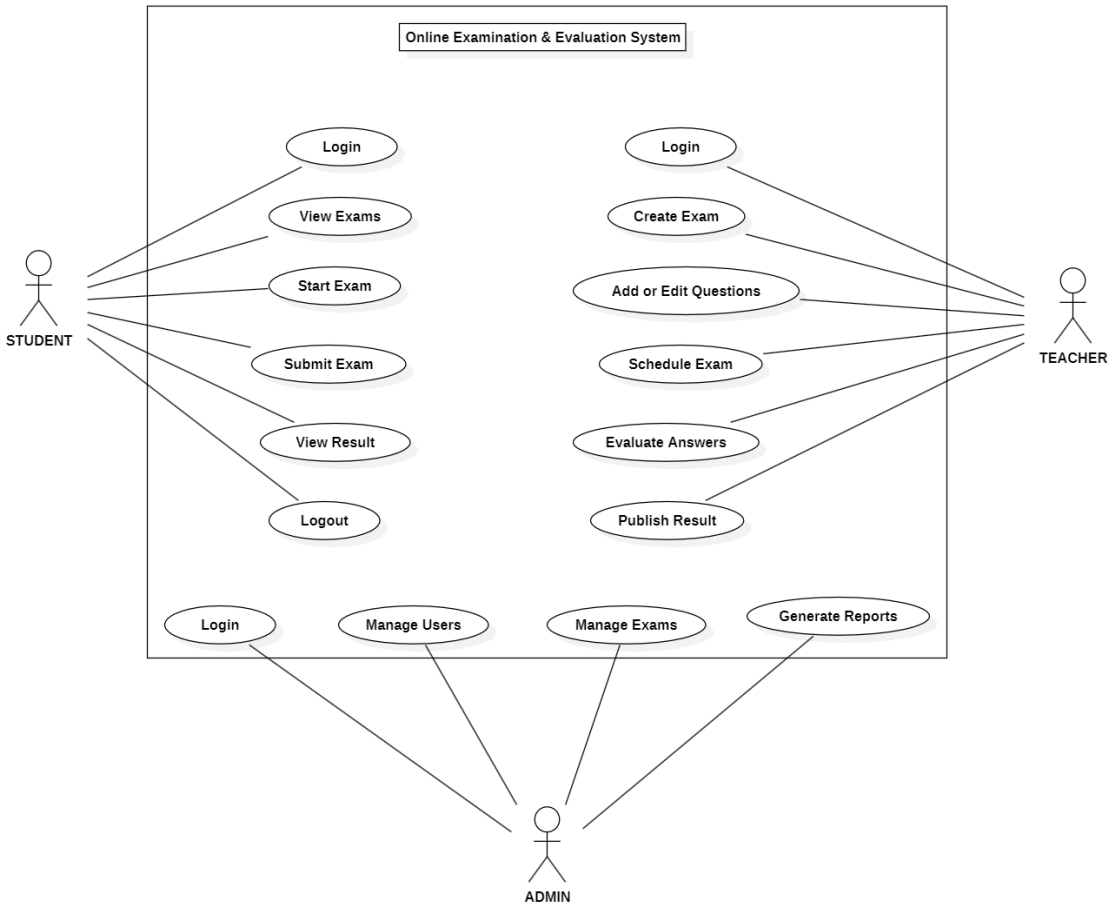


Use Case Diagram:

The Use Case Diagram identifies the functional scope of the system by representing the interactions between users (Student, Teacher, and Admin) and the system functionalities.

Field	Description
Actors	Student, Teacher, Admin
Description	The Use Case Diagram represents the overall functional behavior of the Online Examination & Evaluation System by showing how different users interact with the system to perform operations such as

	login, exam conduction, exam creation, evaluation, and user management.
Data	Login credentials, exam details, questions, answers, results, reports
Stimulus	User initiates a system operation such as login, exam attempt, or exam creation
Response	System performs the requested operation and provides appropriate output



1.5 References

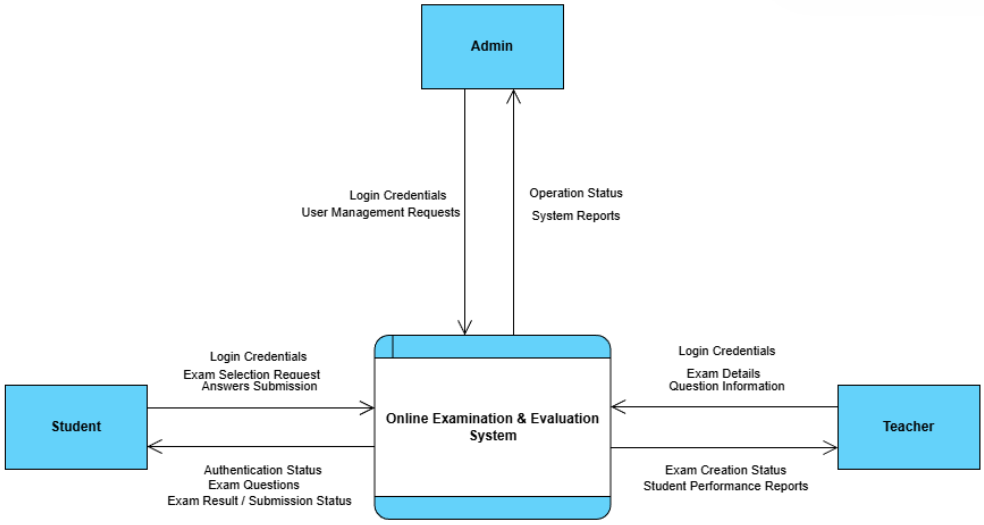
- IEEE Software Requirements Specification (SRS) Standard
- UML 2.x Specification
- Visual Paradigm Documentation

- **Karl E. Wiegers**, *Software Requirements*, 3rd Edition, Microsoft Press, 2013.

2. Overall Description

2.1 Product Perspective

The Online Examination & Evaluation System is a standalone, web-based application that interacts with external entities such as Students, Teachers, and Admins. The system boundary and external interactions are defined using a **Context Diagram**, where the system is represented as a single shown process.



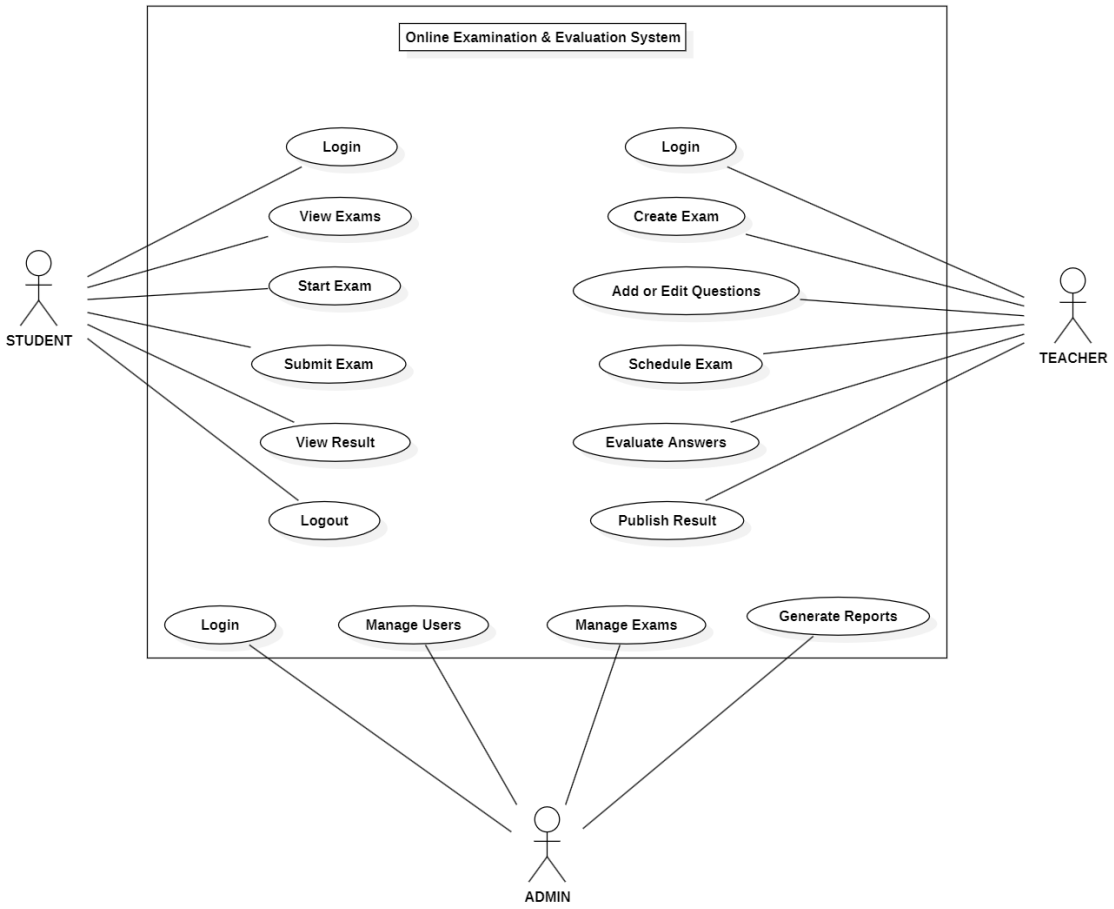
2.2 Product Functions

The major functions of the system include:

- User authentication and authorization
- Online exam scheduling and conduction
- Question management
- Evaluation and result processing
- User and system administration

- Report generation

These system functions are identified and represented using the **Use Case Diagram**.



2.3 User Classes and Characteristics

Student

Students can log in, view available exams, take exams online, submit answers, and view results.

Teacher

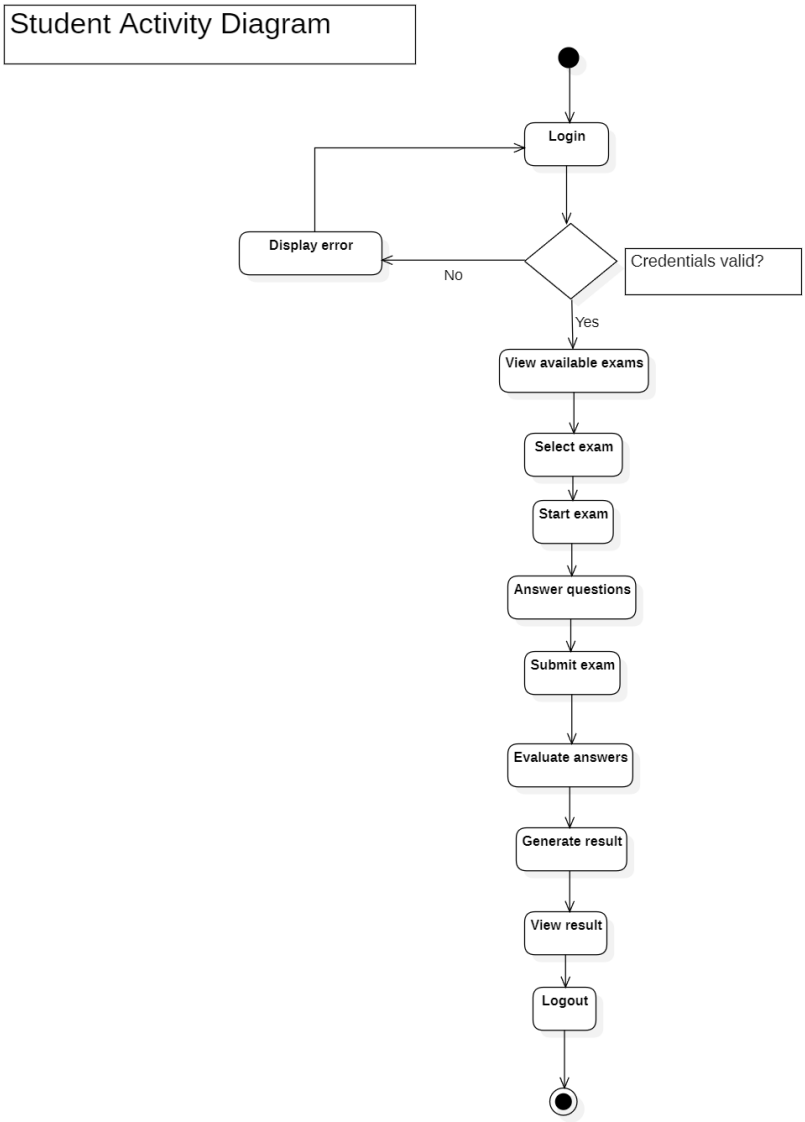
Teachers can log in, create exams, add or edit questions, evaluate student answers, and publish results.

Admin

Admins can manage users, manage exams, and generate system-level reports.

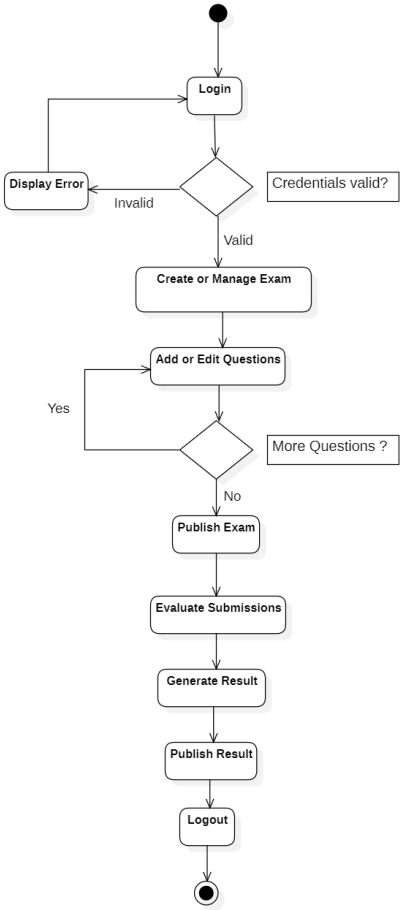
User behavior and workflow are illustrated using **Activity Diagrams**.

Student Activity Diagram:

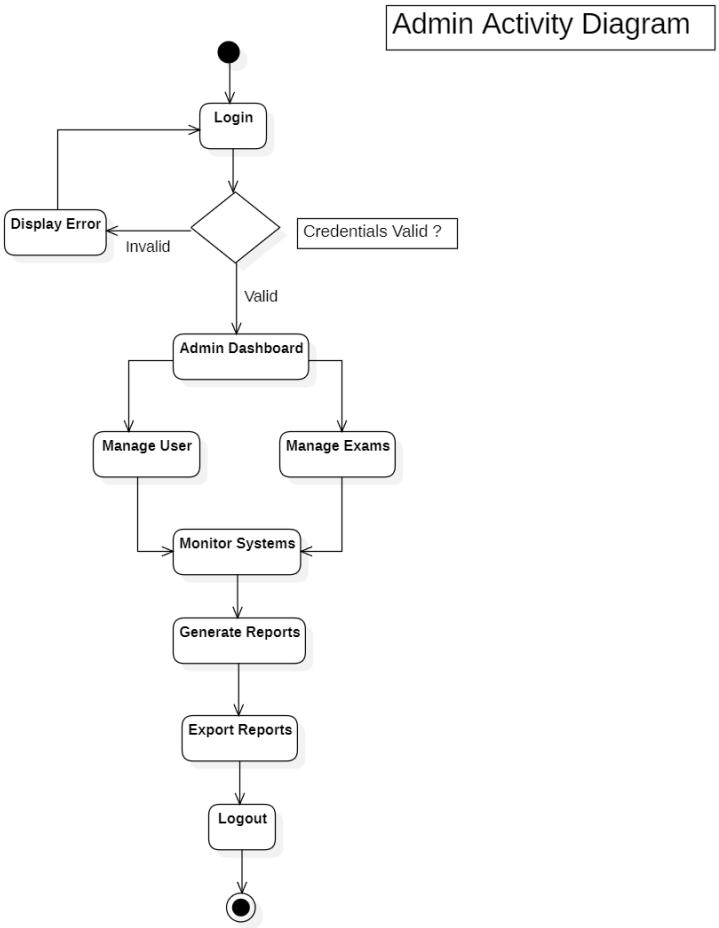


Teacher Activity Diagram:

Teacher Activity Diagram



Admin Activity Diagram:



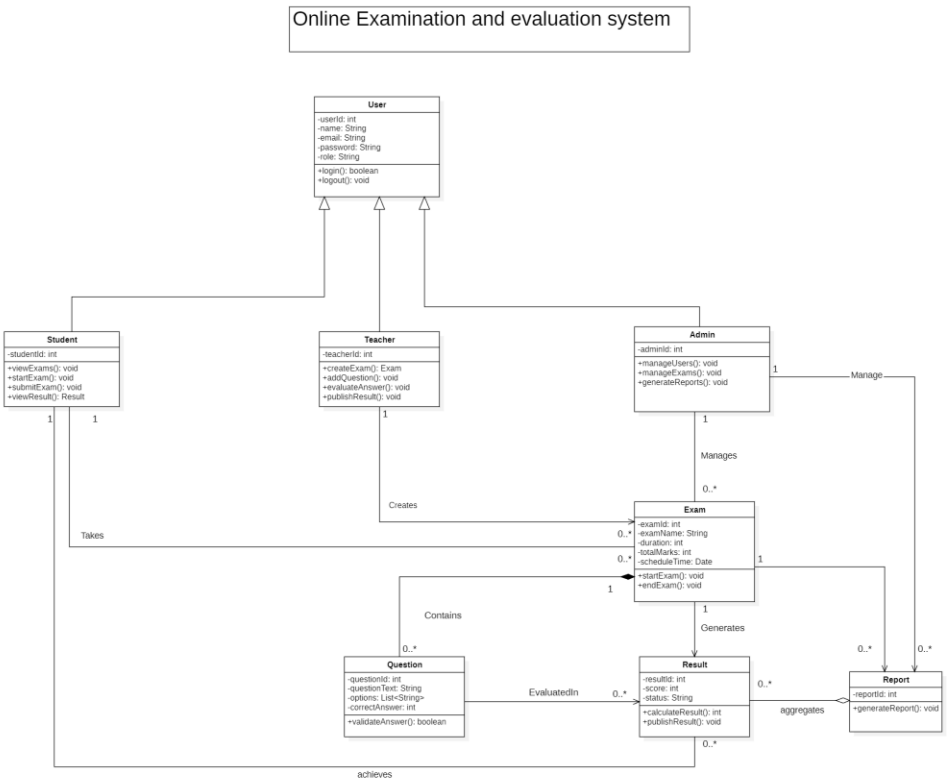
2.4 Operating Environment

- Web-based application
- Accessible via standard web browsers
- Platform-independent

2.5 Design and Implementation Constraints

- UML-based system design
- Role-based access control
- Secure authentication mechanism

Class Diagram:



2.6 User Documentation

The system will provide user guides and help documentation for Students, Teachers, and Admins.

2.7 Assumptions and Dependencies

The following assumptions and dependencies are considered during the development and specification of the Online Examination & Evaluation System. These factors may influence system behavior, performance, or implementation if they change.

Assumptions

1. All users (Students, Teachers, and Admins) are registered in the system before accessing system functionalities.
2. Users possess basic knowledge of operating web-based applications.
3. The system will be accessed through a supported web browser.

4. Internet connectivity is available throughout the examination process.
5. Examinations are scheduled and managed by authorized Teachers or Admins.

Dependencies

1. The system depends on the availability of a web server and backend database server.
2. Proper functioning of the system depends on continuous network connectivity.
3. User authentication depends on the validity of stored user credentials.
4. Examination and result processing depend on accurate data storage and retrieval.
5. System availability depends on server uptime and maintenance schedules.

3. External Interface Requirements

3.1 User Interfaces

The system provides separate interfaces for each user role. Each interface supports role-specific operations such as exam conduction, exam creation, and system administration.

3. External Interface Requirements

This section describes the interfaces between the Online Examination & Evaluation System and external entities, including users, software components, and communication mechanisms. The interface requirements are derived from the **Context Diagram, DFD Level-1, Activity Diagrams, and Sequence Diagrams**.

3.1 User Interfaces

The system provides role-based user interfaces to support different categories of users, namely Students, Teachers, and Admins. Each interface is designed to provide access only to the functionalities permitted for the respective role.

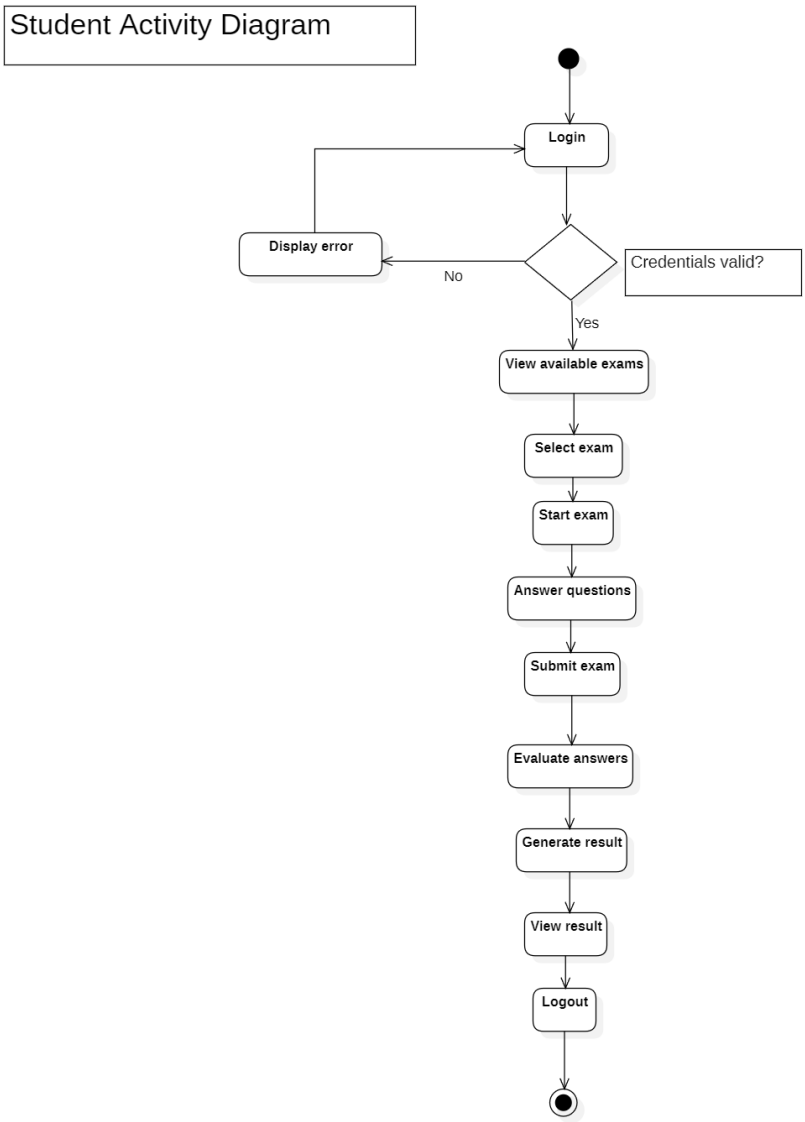
Student Interface

The student interface allows students to:

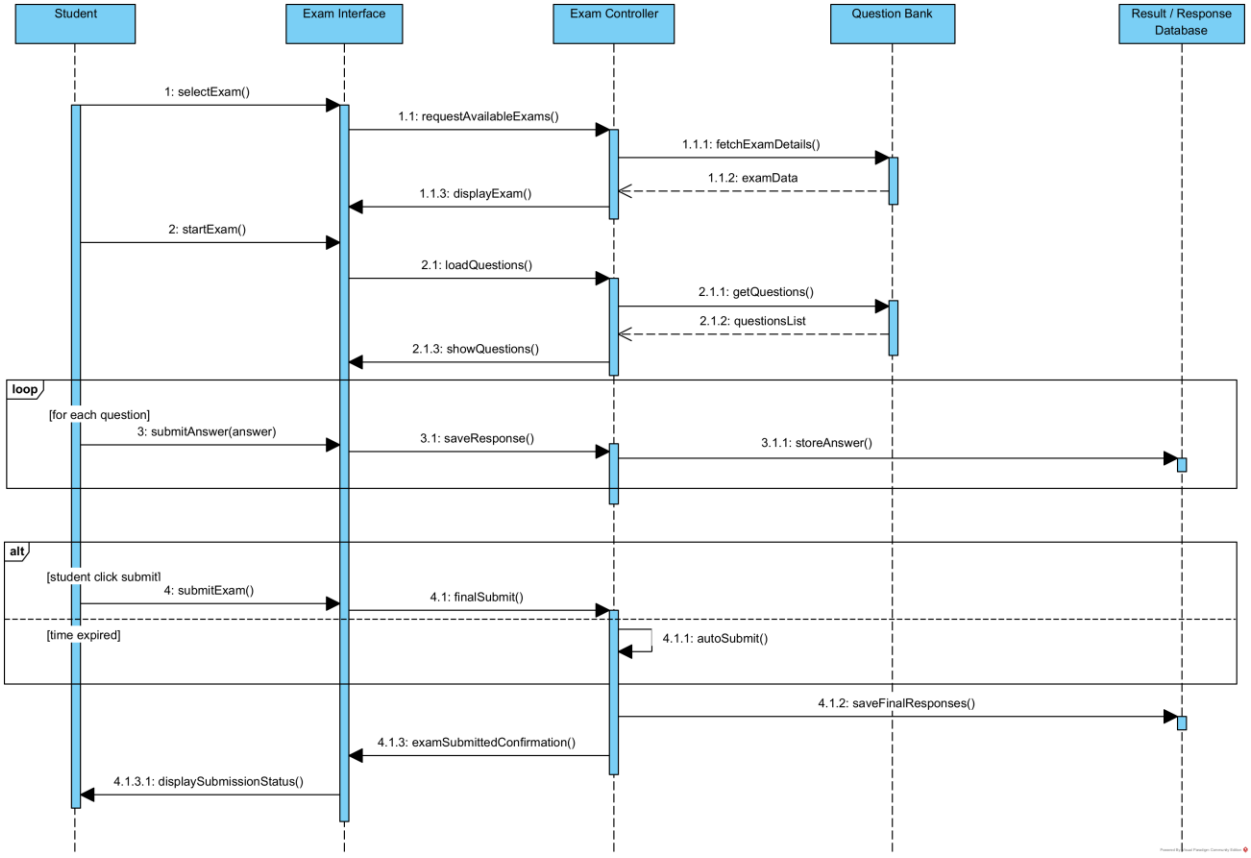
- Log in to the system
- View available examinations
- Attempt examinations online
- Submit answers
- View examination results

The workflow of student interaction is represented using the **Student Activity Diagram**, while detailed interactions are illustrated in the **Sequence Diagram for Taking an Exam**.

Student Activity Diagram:



Taking an Exam Sequence Diagram:



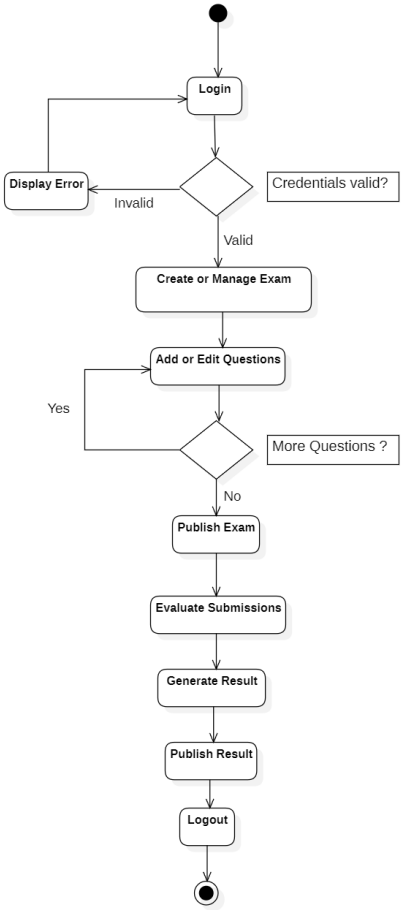
Teacher Interface

The teacher interface enables teachers to:

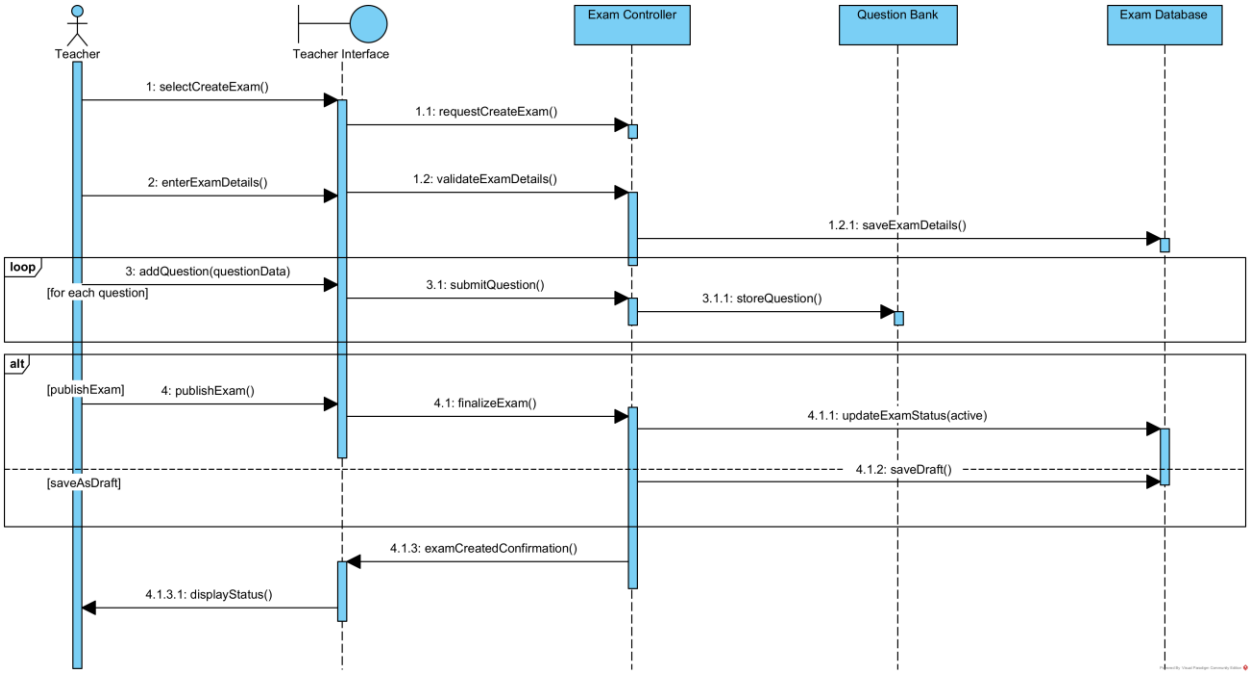
- Log in securely
- Create and manage examinations
- Add, update, or delete questions
- Evaluate student responses
- Publish examination results

Teacher workflows and interactions are represented using the **Teacher Activity Diagram** and the **Sequence Diagram for Exam Creation**.

Teacher Activity Diagram



Teacher Sequence Diagram :



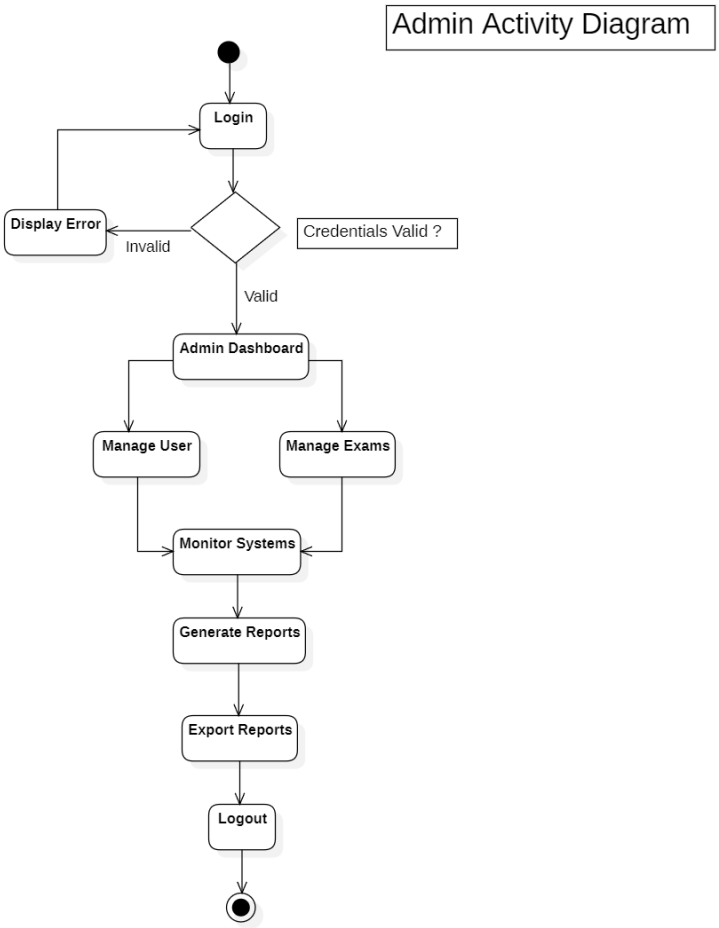
Admin Interface

The admin interface provides administrative control over the system, allowing admins to:

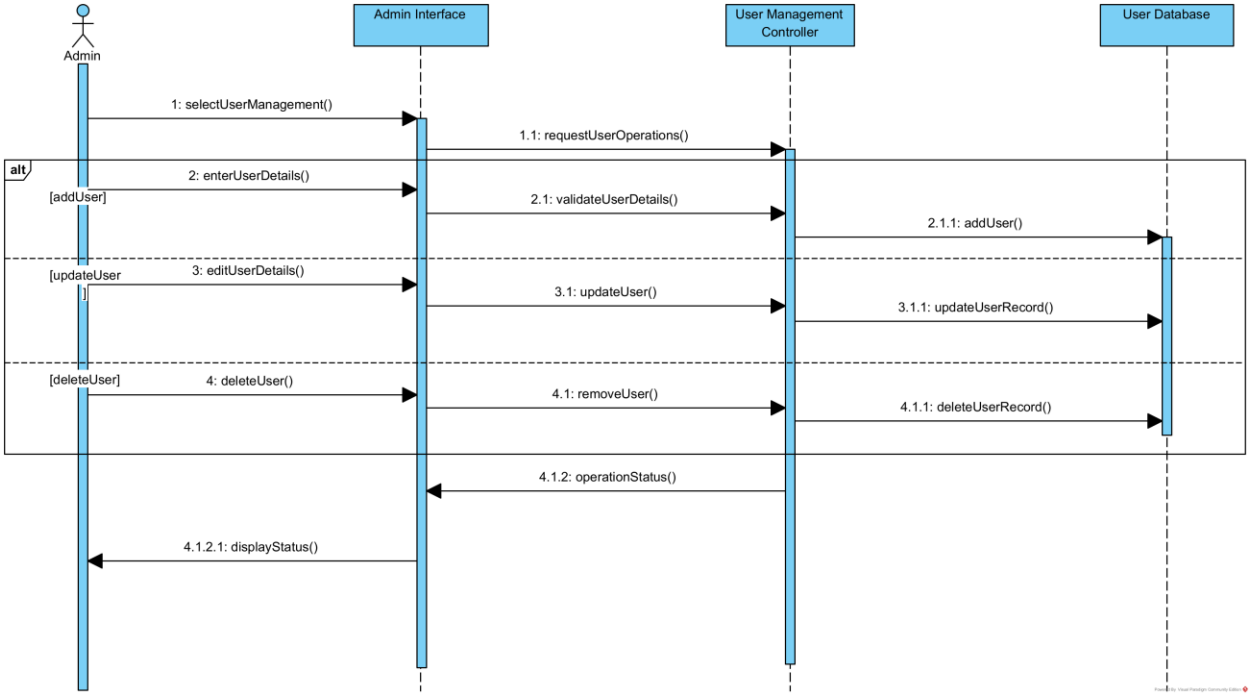
- Manage user accounts
- Manage examinations
- Generate reports
- Monitor system activities

Administrative workflows are illustrated using the **Admin Activity Diagram** and the **Sequence Diagram for User Management**.

Admin activity Diagram:



Admin User Management diagram:



3.2 Hardware Interfaces

The Online Examination & Evaluation System does not require any specialized hardware. The system can operate on standard computing devices such as desktops, laptops, and mobile devices with internet connectivity.

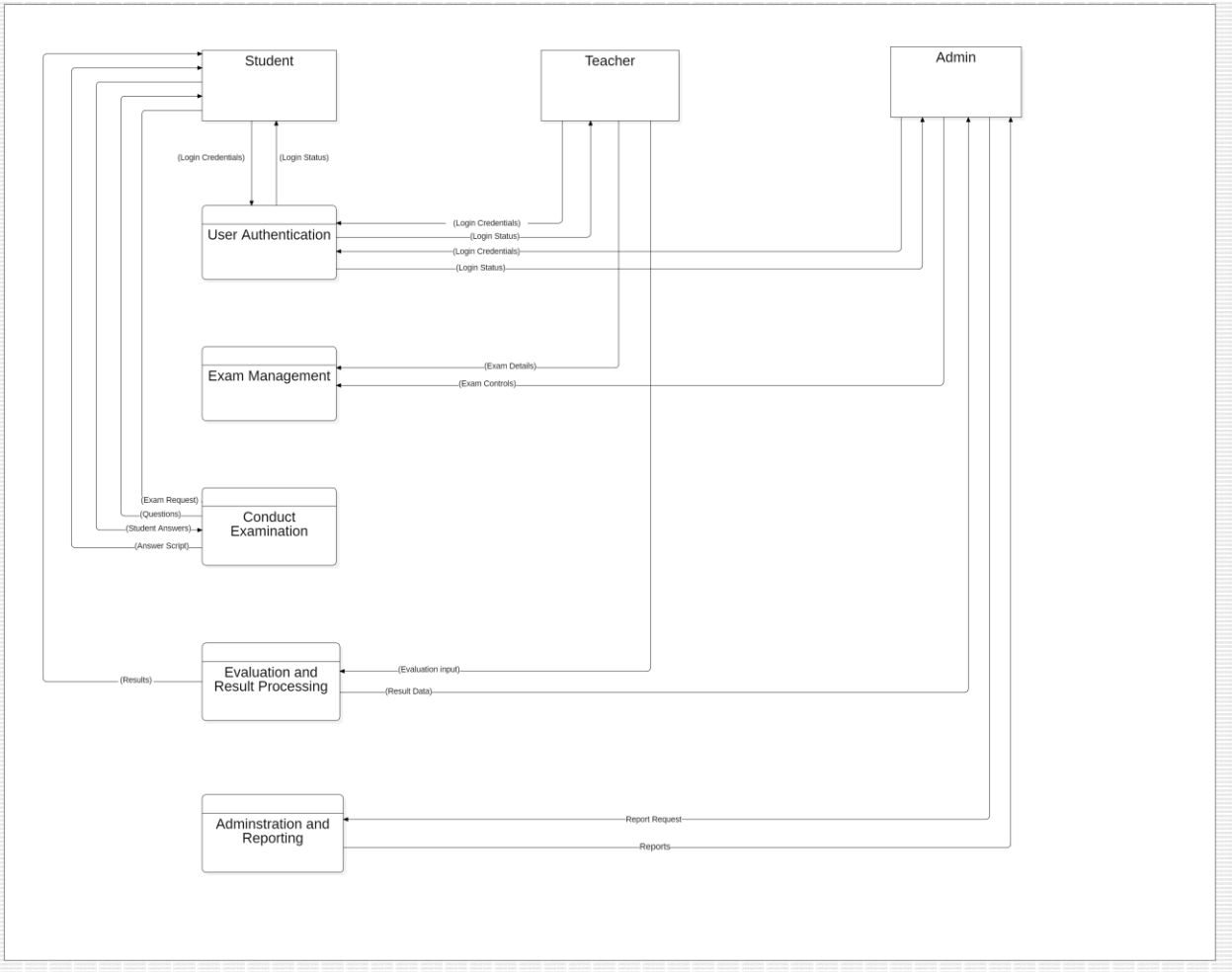
3.3 Software Interfaces

The system interacts with various software components to perform its operations. These include:

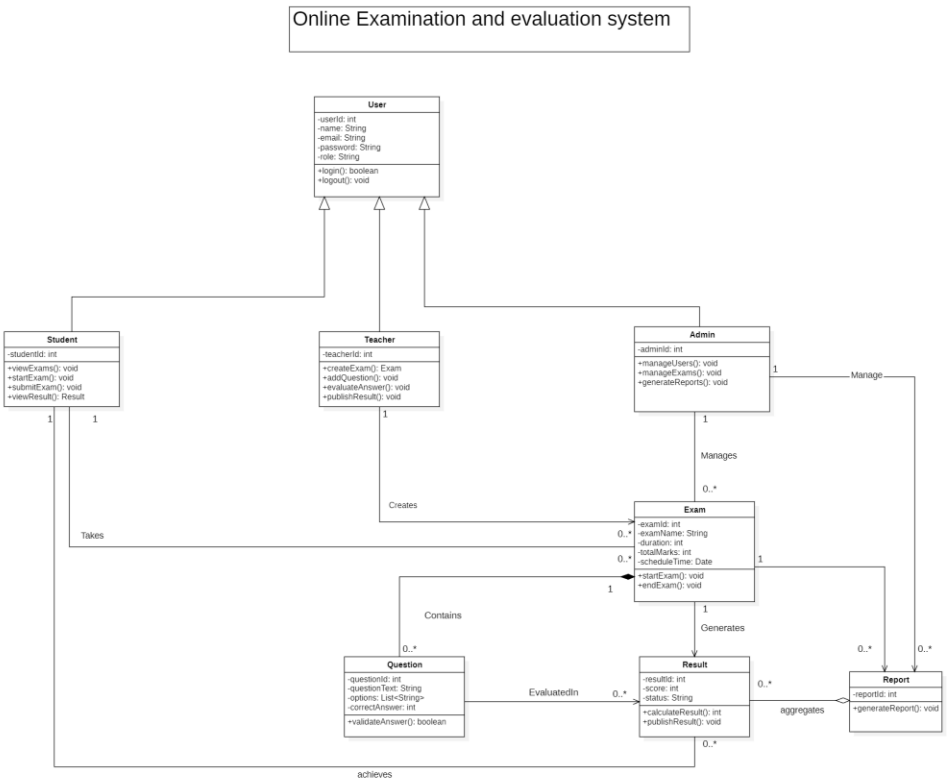
- A backend database system for storing user information, examination data, responses, and results
- A web server for handling client requests and responses

The interaction between software components and data flow is represented using **DFD Level-1** and the **Class Diagram**, which defines the structural relationships among system entities.

DFD Level 1:



Class Diagram:



3.4 Communication Interfaces

The system uses standard web-based communication protocols such as HTTP and HTTPS for client-server communication. Secure communication mechanisms are assumed to protect sensitive data such as login credentials and examination responses.

4. System Features

This section describes the major system features and functional requirements. These features are identified from the **Use Case Diagram**, **Activity Diagrams**, **Sequence Diagrams**, and **DFD Level-1**.

4.1 User Authentication

Description

The system provides a secure authentication mechanism that verifies user credentials before granting access. Upon successful authentication, users are redirected to their respective dashboards based on their roles.

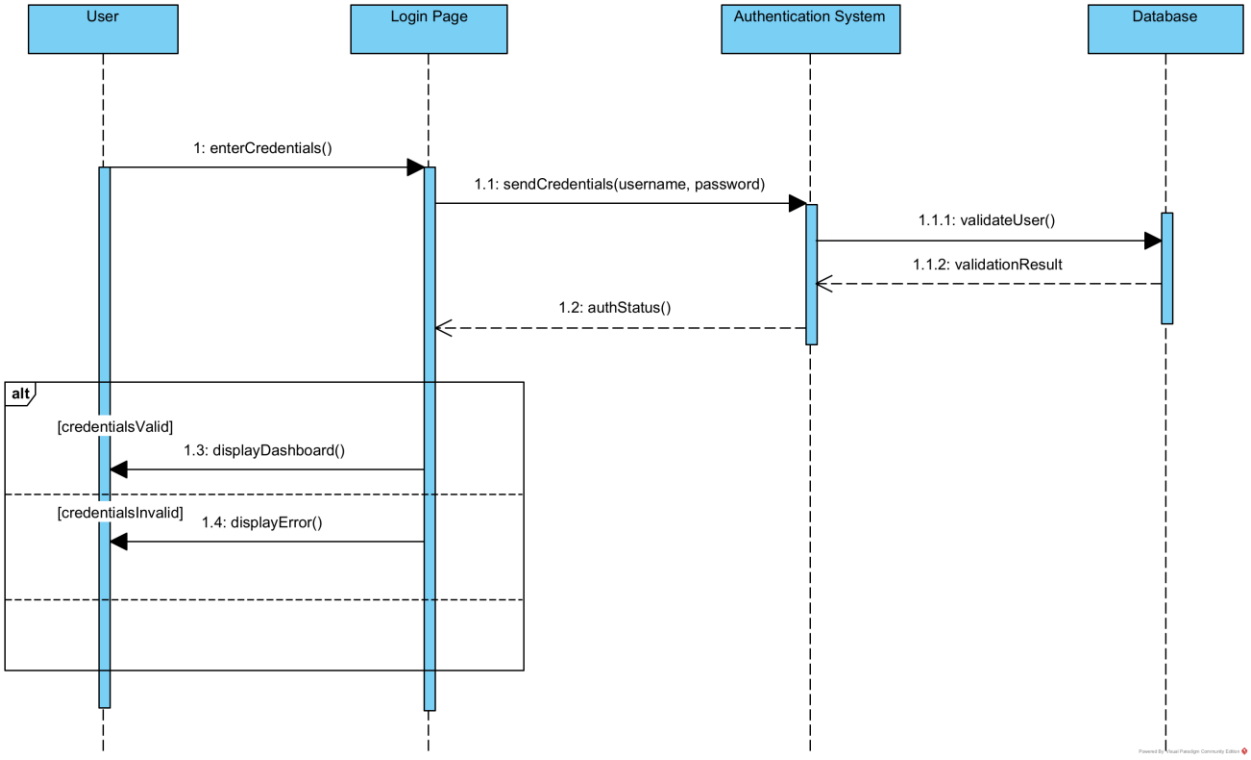
Priority

High

Stimulus/Response Sequence

- User enters login credentials
- System validates credentials
- System grants or denies access

User Authentication Sequence diagram:



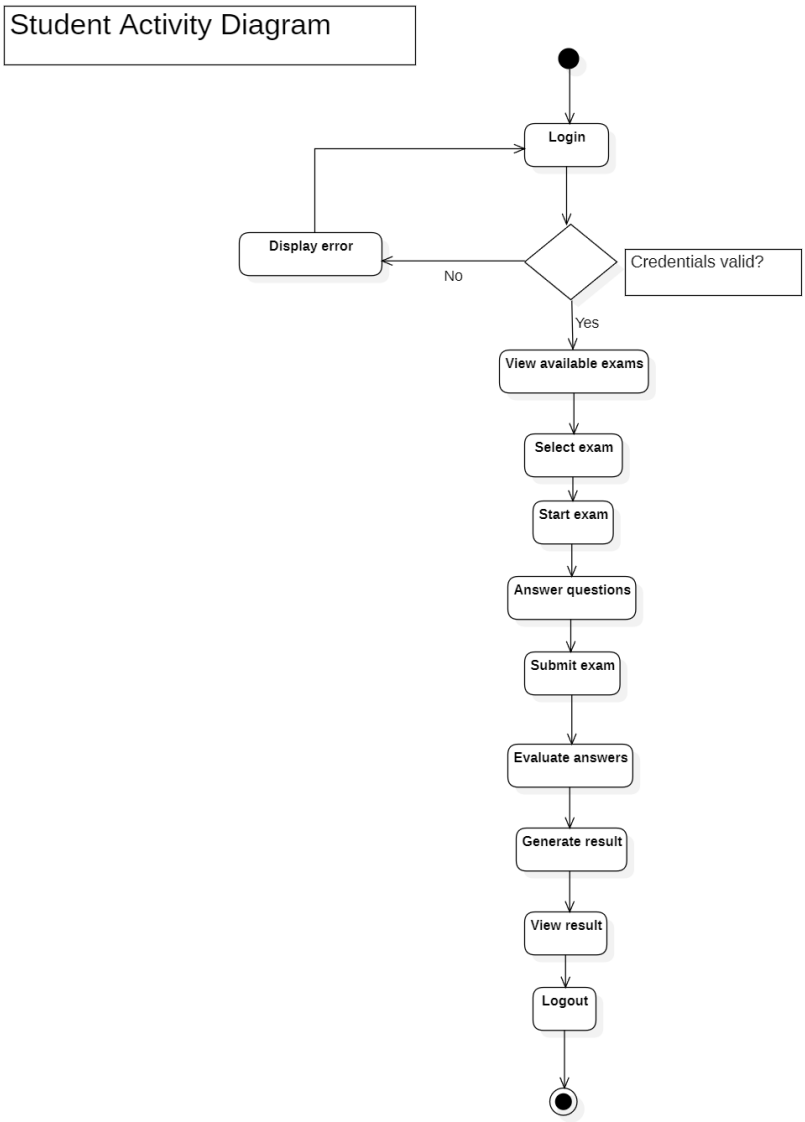
4.12 Online Examination

Description

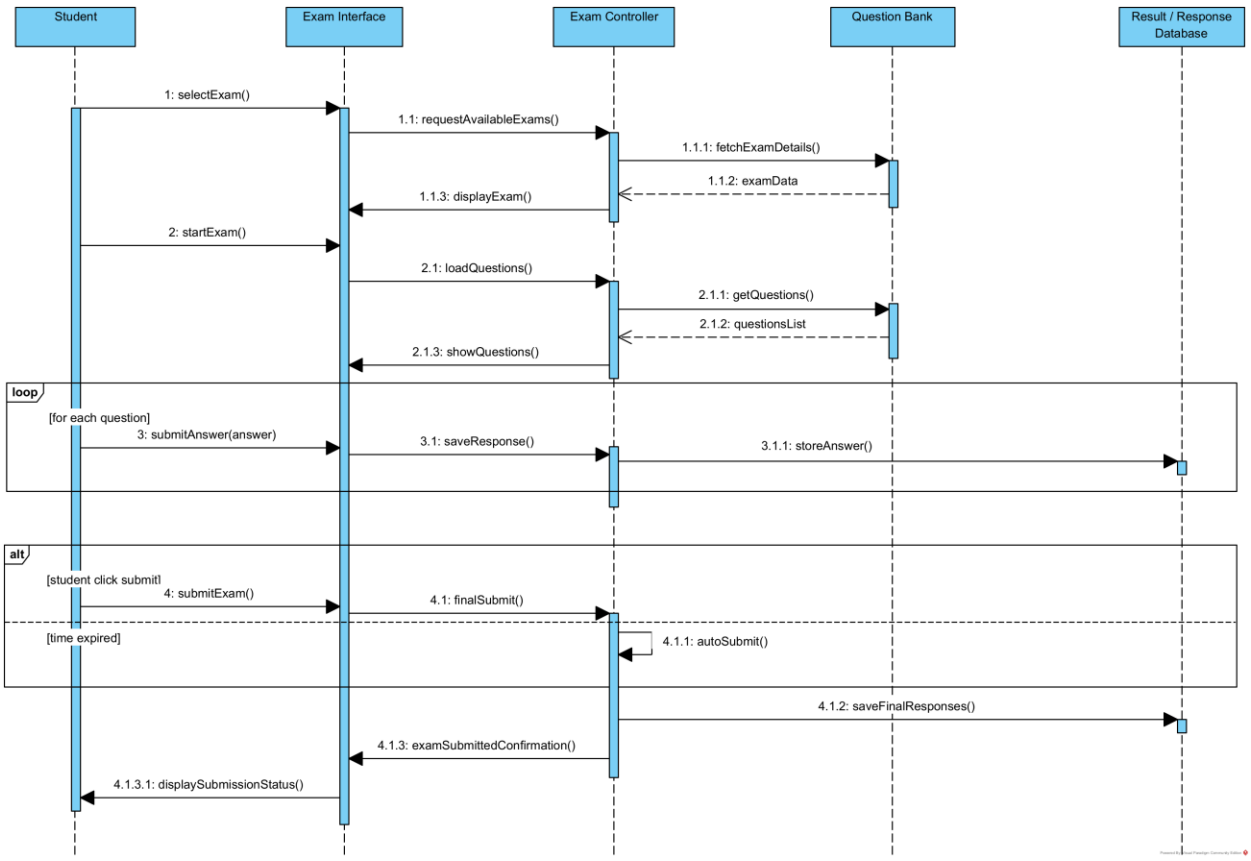
The system enables students to take examinations online by selecting an available exam, answering questions, and submitting responses. The system automatically records answers and updates submission status.

Stimulus/Response Sequence

- Student selects an exam
- System displays questions
- Student submits answers
- System confirms submission



Taking Exam Sequence Diagram:



4.13 Exam Management

Description

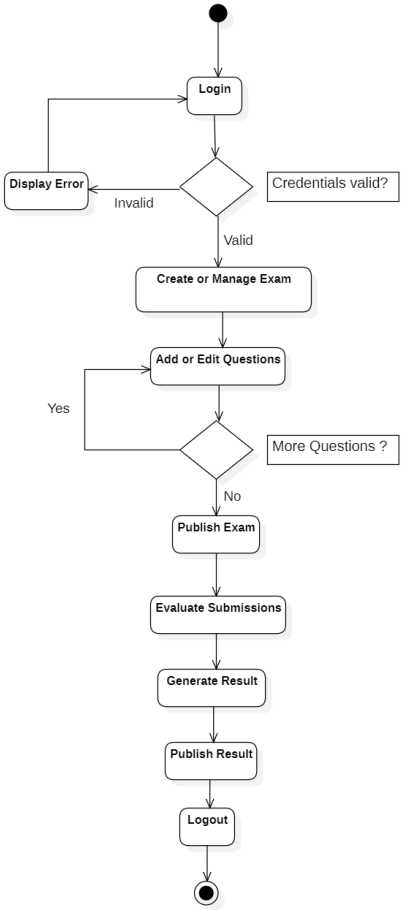
The system allows teachers to create and manage examinations. Teachers can define exam details, add questions, evaluate answers, and publish results.

Stimulus/Response Sequence

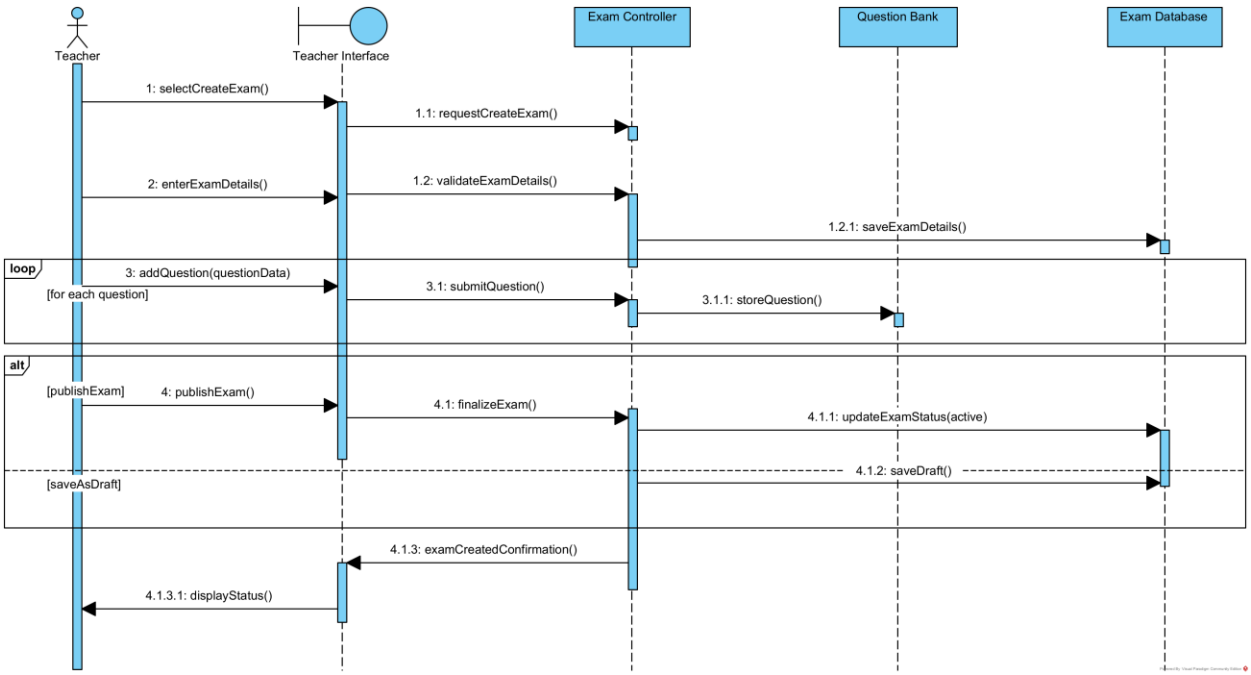
- Teacher creates exam
- Teacher adds questions
- System stores exam details
- Teacher publishes exam

Teacher Activity:

Teacher Activity Diagram



Taking Exam:



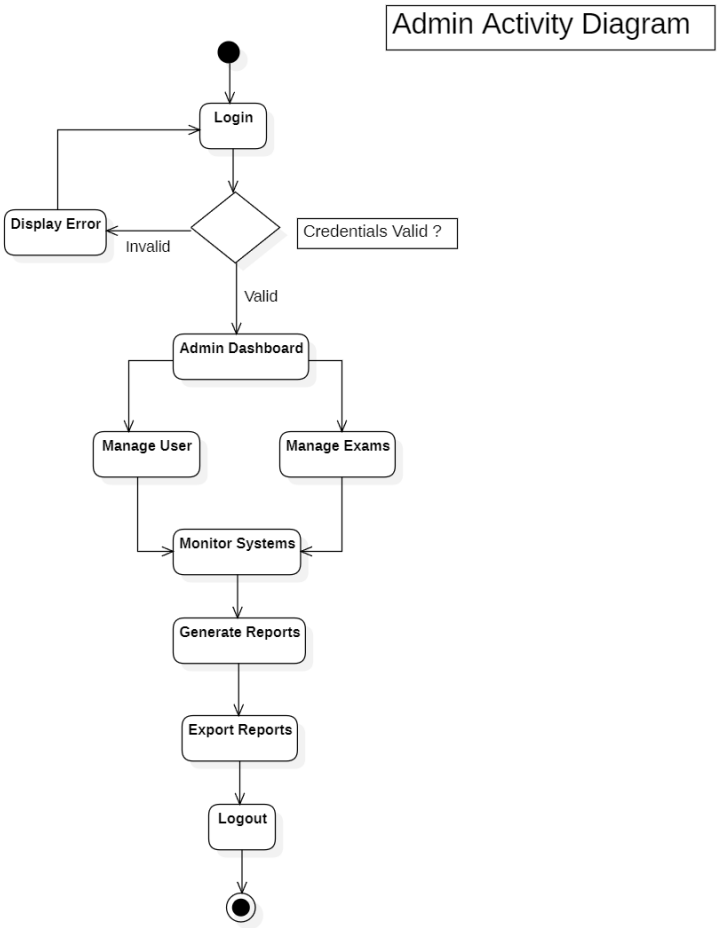
4.14 User Management

Description

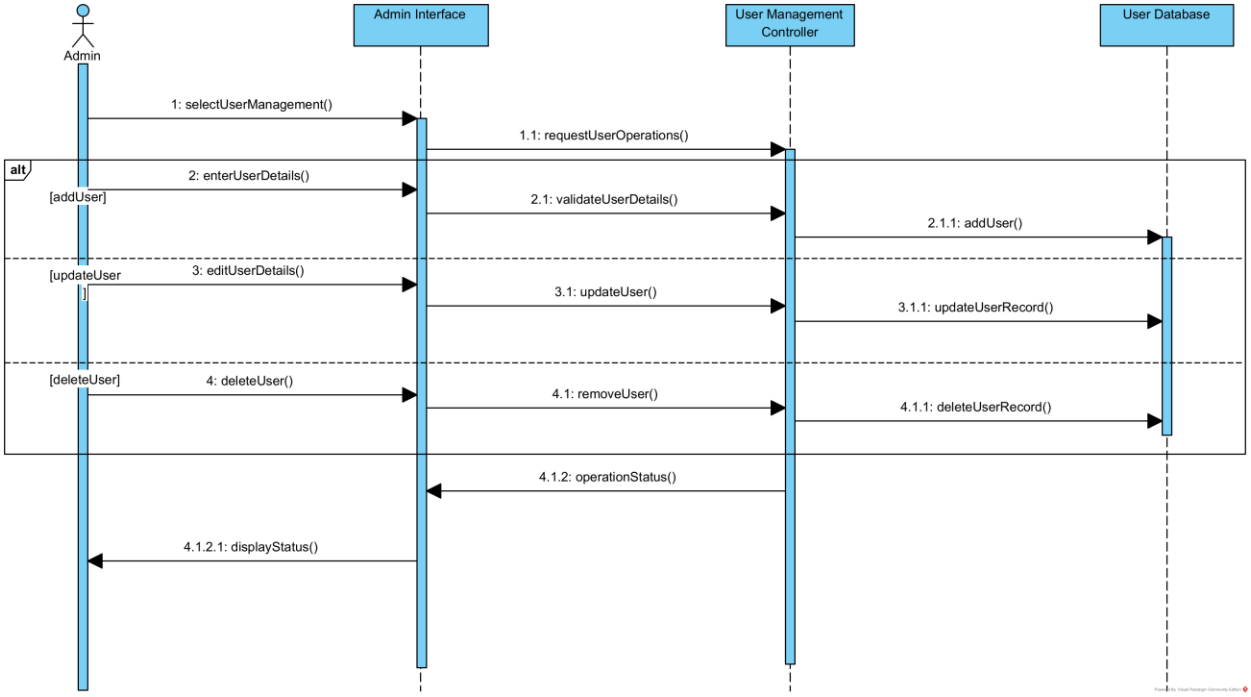
The system allows admins to manage users by adding, updating, or deleting accounts. Admins can also generate system reports and monitor activities.

Stimulus/Response Sequence

- Admin selects user operation
- System processes request
- System updates database
- System displays status



Admin sequence Diagram:



5. Non-Functional Requirements

5.1 Performance Requirements

- The system should support multiple users simultaneously.
- System response time should be minimal during examination submission.

5.2 Security Requirements

- Authentication is mandatory for all users.
- Role-based access control must be enforced.
- Examination data must be protected against unauthorized access.

5.3 Reliability Requirements

- The system should ensure data integrity.
- Examination responses should not be lost during submission.

5.4 Usability Requirements

- Interfaces should be simple and intuitive.
- The system should be easy to navigate for all users.

5.5 Business Rules

- Only teachers can create and manage exams.
- Only admins can manage user accounts.
- Students can only attempt exams assigned to them.

6. Other Requirements

- Accurate result evaluation
- Secure data storage
- Proper report generation

Appendix A: Glossary

Term	Description
Admin	A system user responsible for managing users, exams, and system reports.
Activity Diagram	A UML diagram that represents the workflow of activities within the system.
Class Diagram	A UML diagram that shows system classes, attributes, and relationships.
Context Diagram	A Level-0 DFD that represents the system as a single process and its interaction with external entities.
DFD	Data Flow Diagram, used to represent the flow of data within the system.
DFD Level-0	A high-level DFD showing system interactions with external entities.
DFD Level-1	A detailed DFD showing internal system processes and data flows.
Exam	An online assessment conducted using the system.
Sequence Diagram	A UML diagram showing interaction between objects in a time sequence.
Student	A user who attempts examinations and views results.

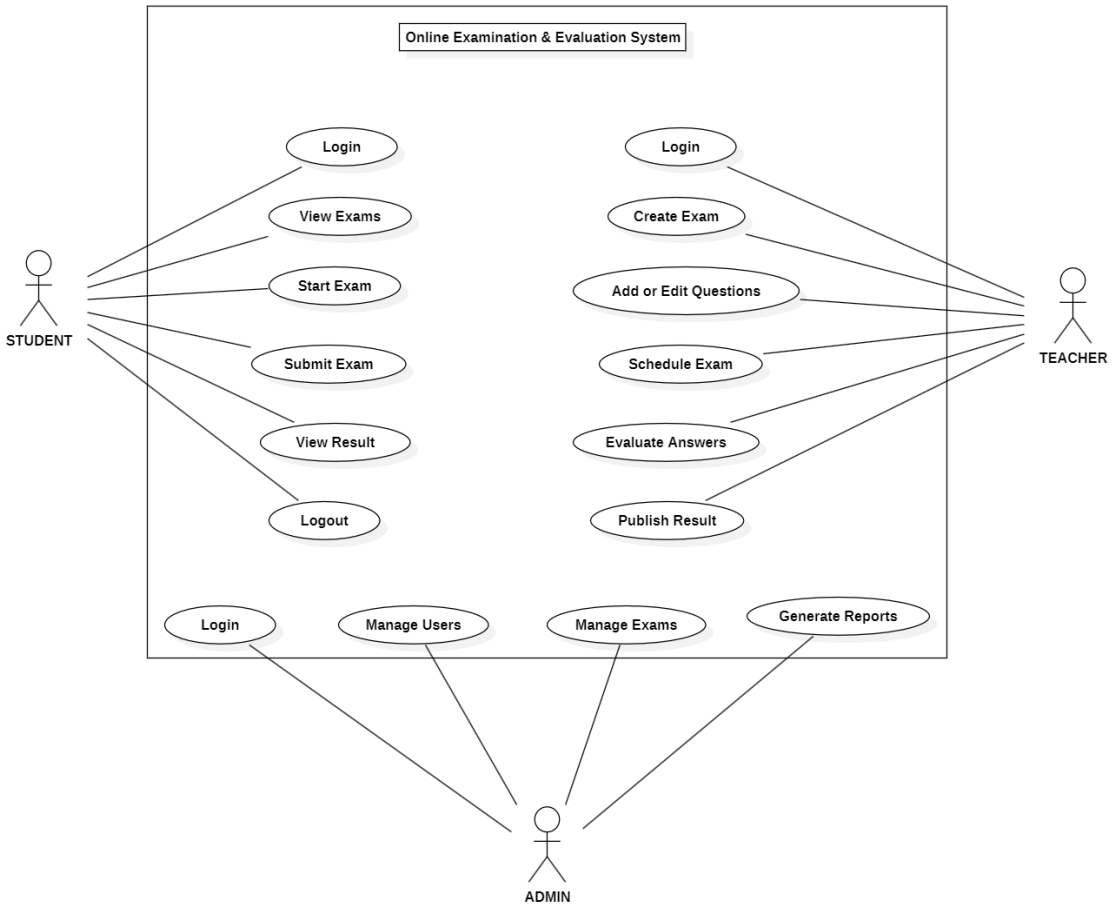
Teacher	A user who creates exams, manages questions, and evaluates results.
UML	Unified Modeling Language used for system modeling.

Appendix B: Analysis Models

This section presents the analysis and design models used to understand, specify, and validate the requirements of the Online Examination & Evaluation System. The following diagrams are included to represent the system from different perspectives.

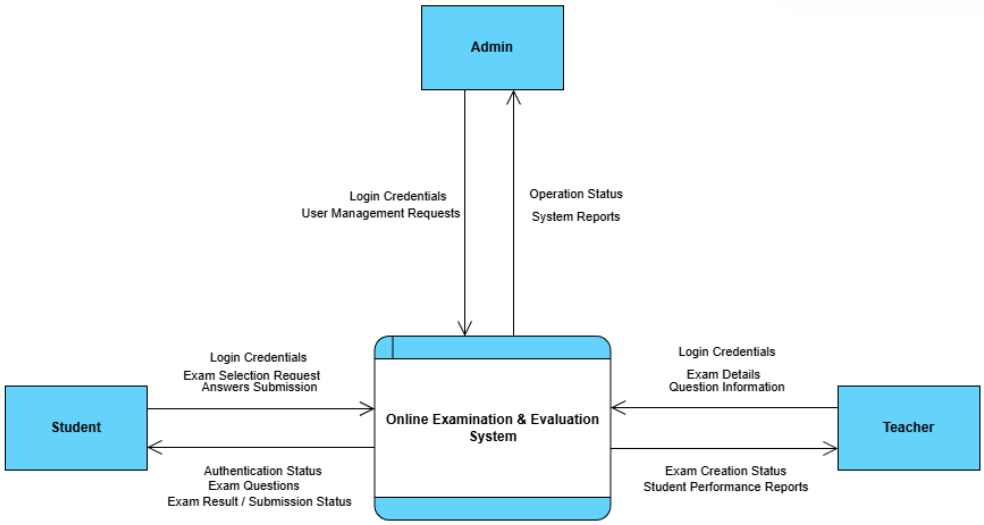
B.1 Use Case Diagram

The Use Case Diagram illustrates the functional requirements of the system and identifies interactions between the system and external actors such as Student, Teacher, and Admin.



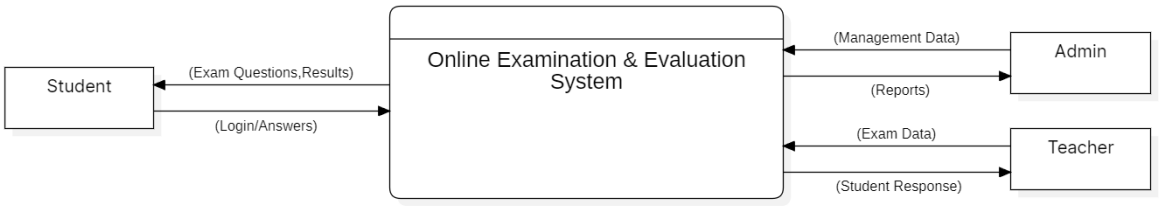
B.2 Context Diagram

The Context Diagram represents the system as a single process and shows its interaction with external entities, defining the system boundary.



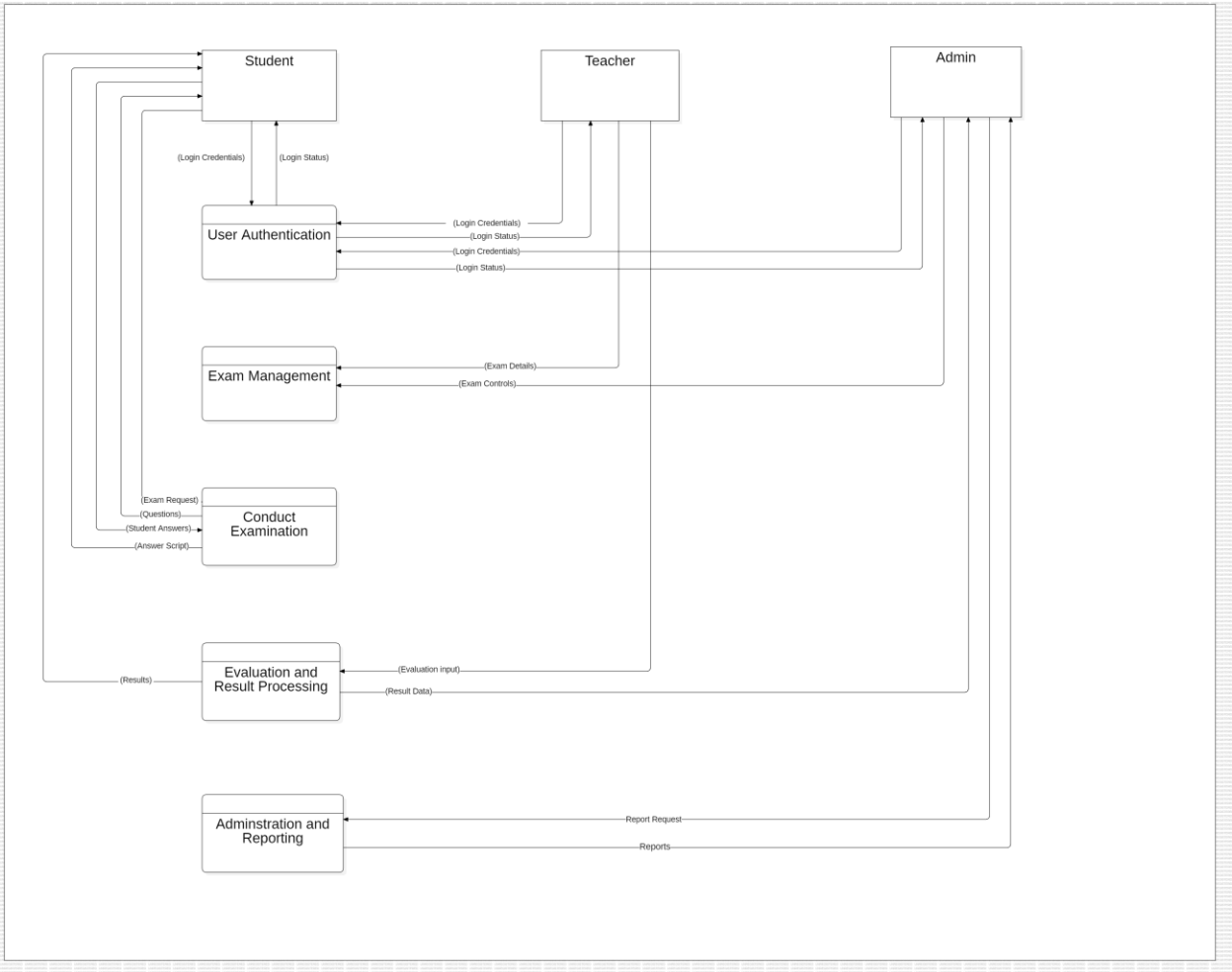
B.3 Data Flow Diagram – Level 0

The DFD Level-0 provides a high-level overview of data flow between the system and external entities without showing internal processing details.



B.4 Data Flow Diagram – Level 1

The DFD Level-1 expands the system into major processes and illustrates detailed data flow between processes, data stores, and external entities.

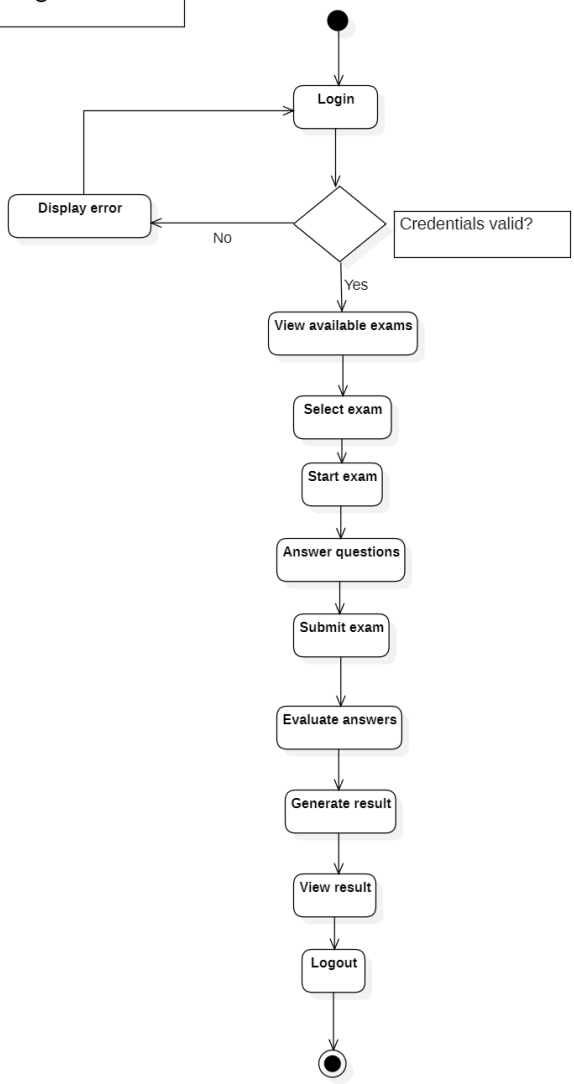


B.5 Activity Diagrams

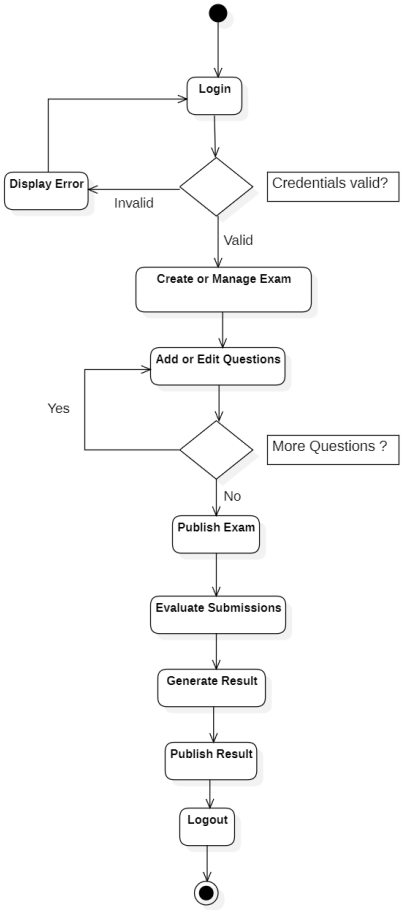
Activity Diagrams describe the workflow of activities performed by different users.

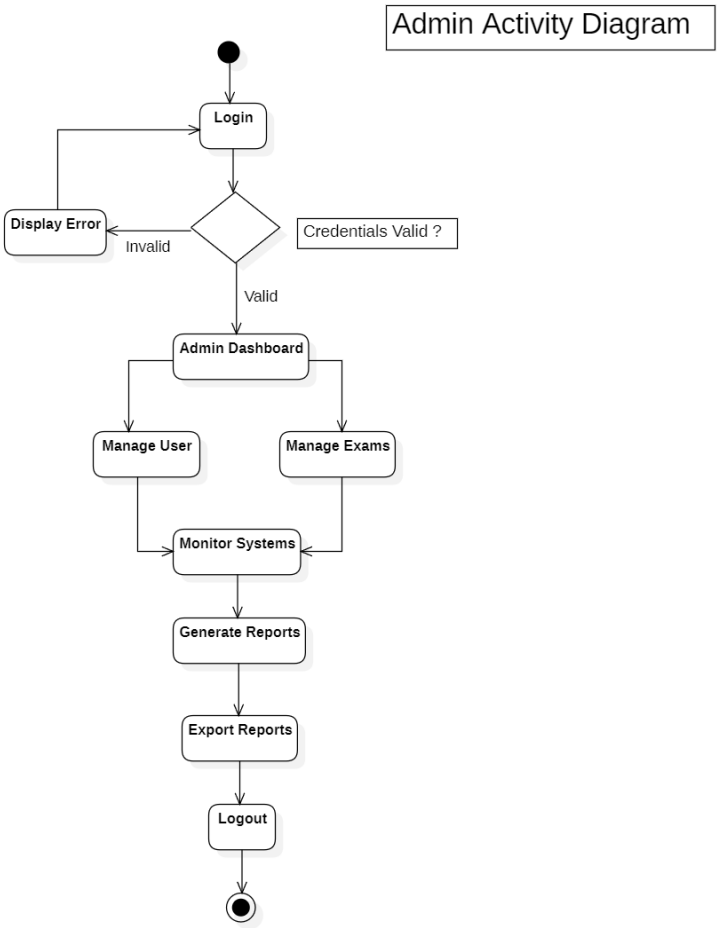
- **Student Activity Diagram**
- **Teacher Activity Diagram**
- **Admin Activity Diagram**

Student Activity Diagram



Teacher Activity Diagram



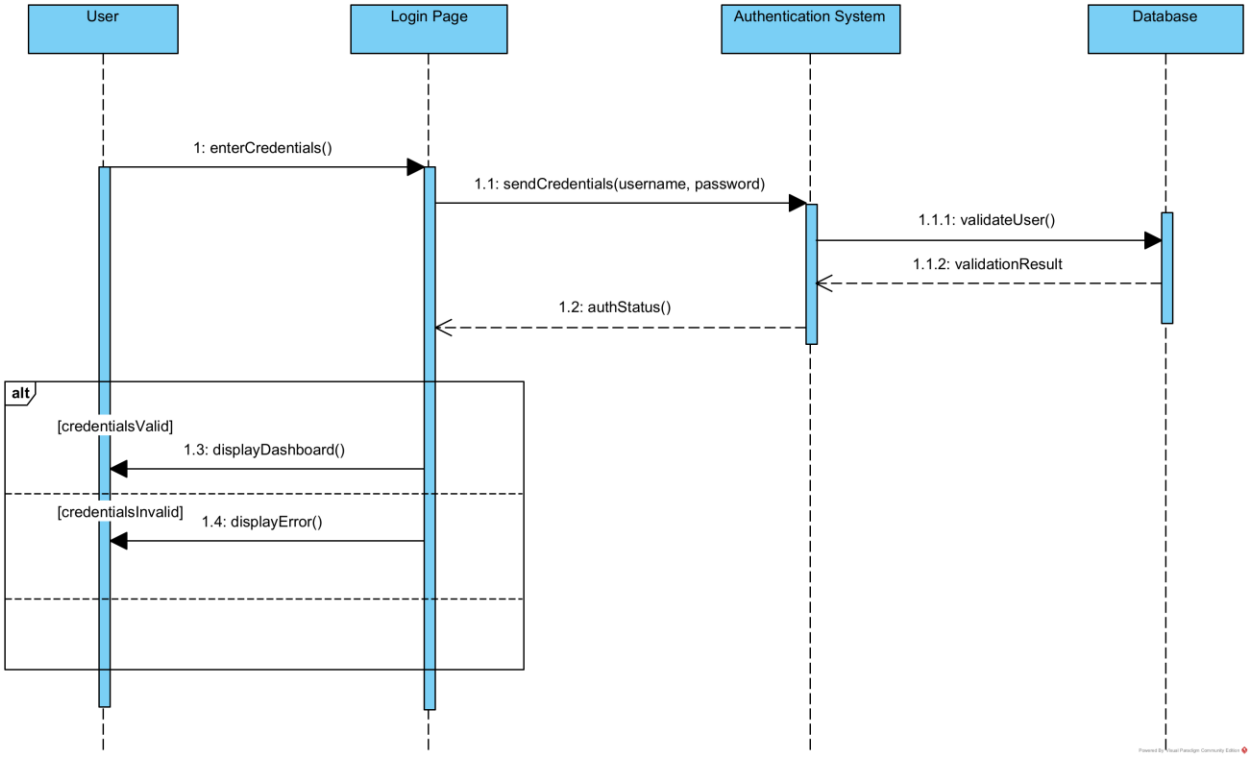


B.6 Sequence Diagrams

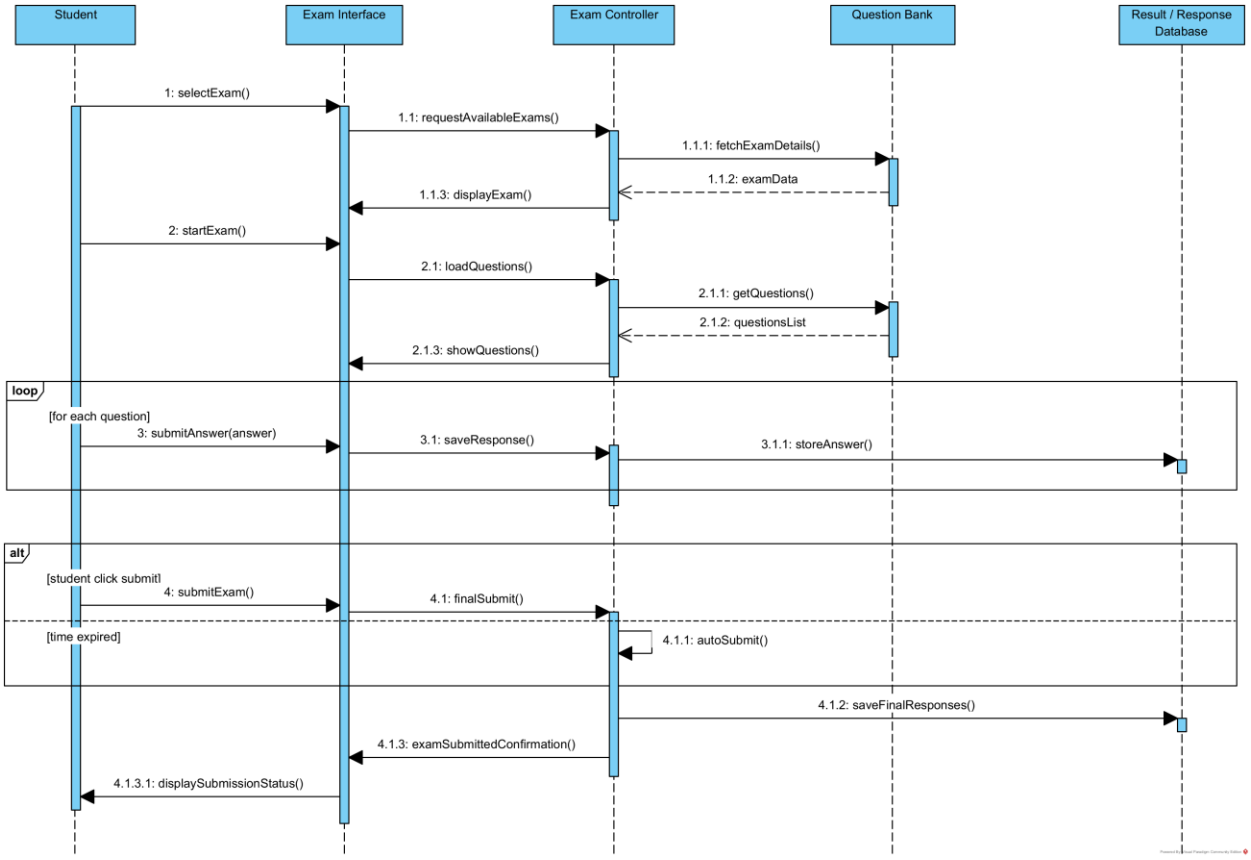
Sequence Diagrams illustrate the interaction between system components over time for key system operations.

- User Authentication
- Taking an Exam (Student)
- Creating an Exam (Teacher)
- User Management (Admin)

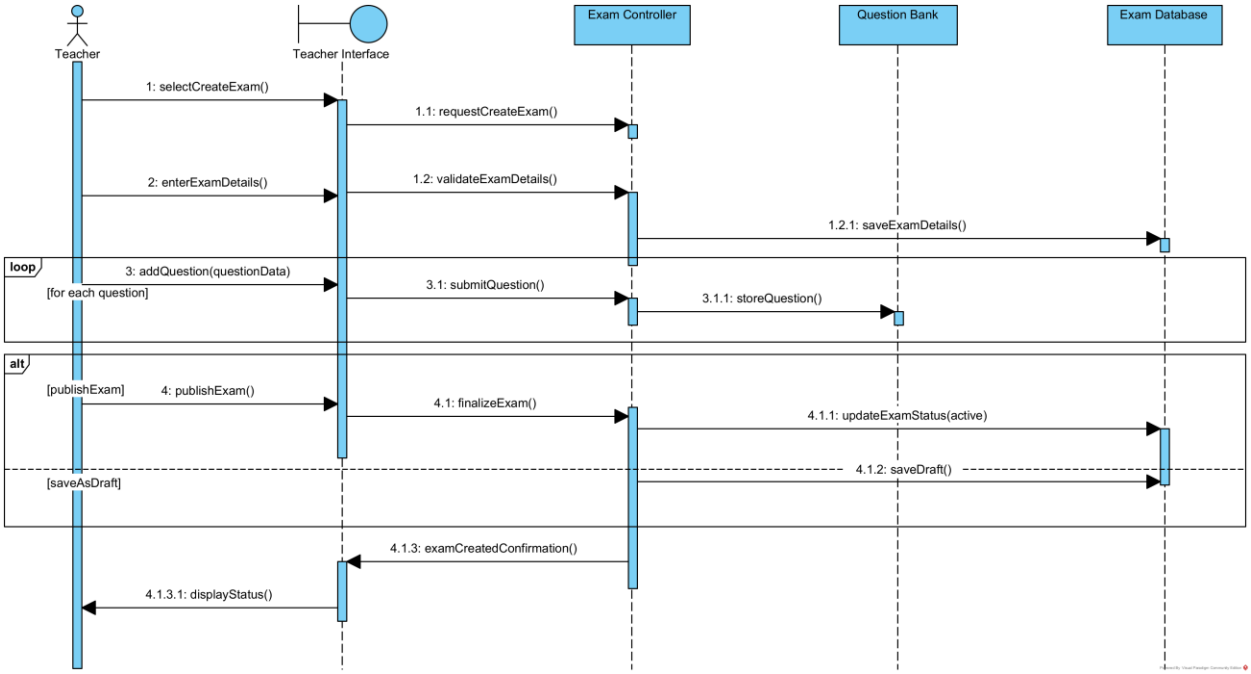
1.User Authentication



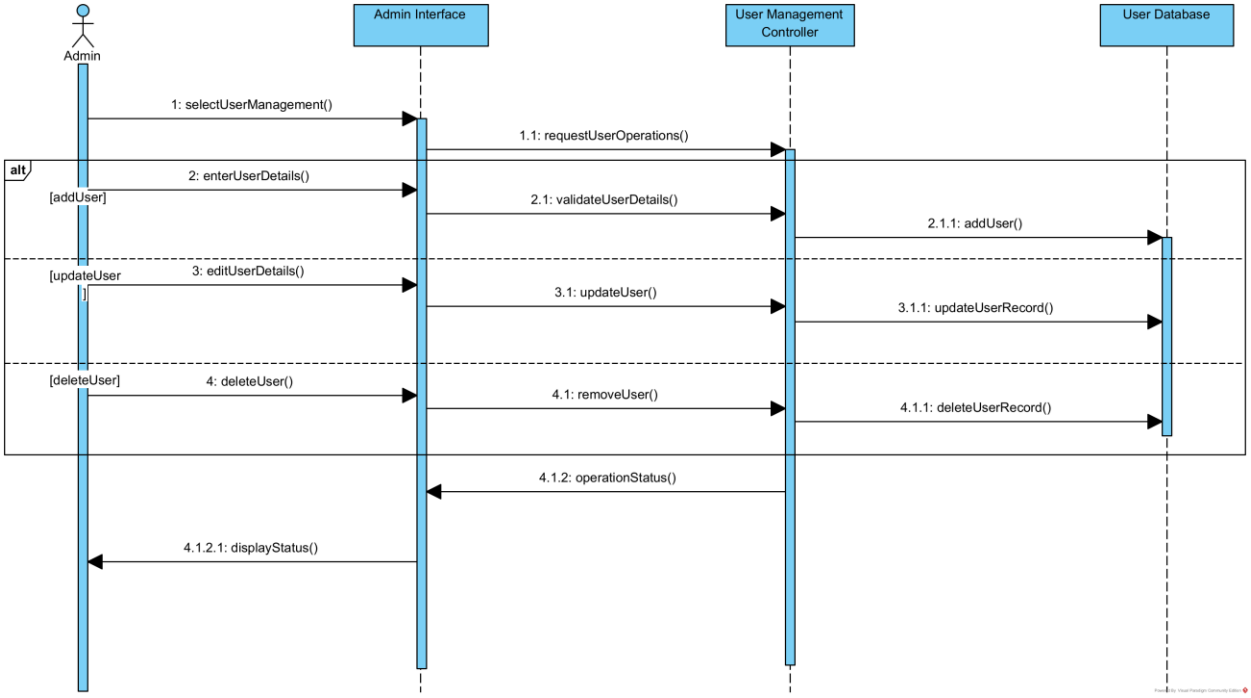
2.Taking an Exam:

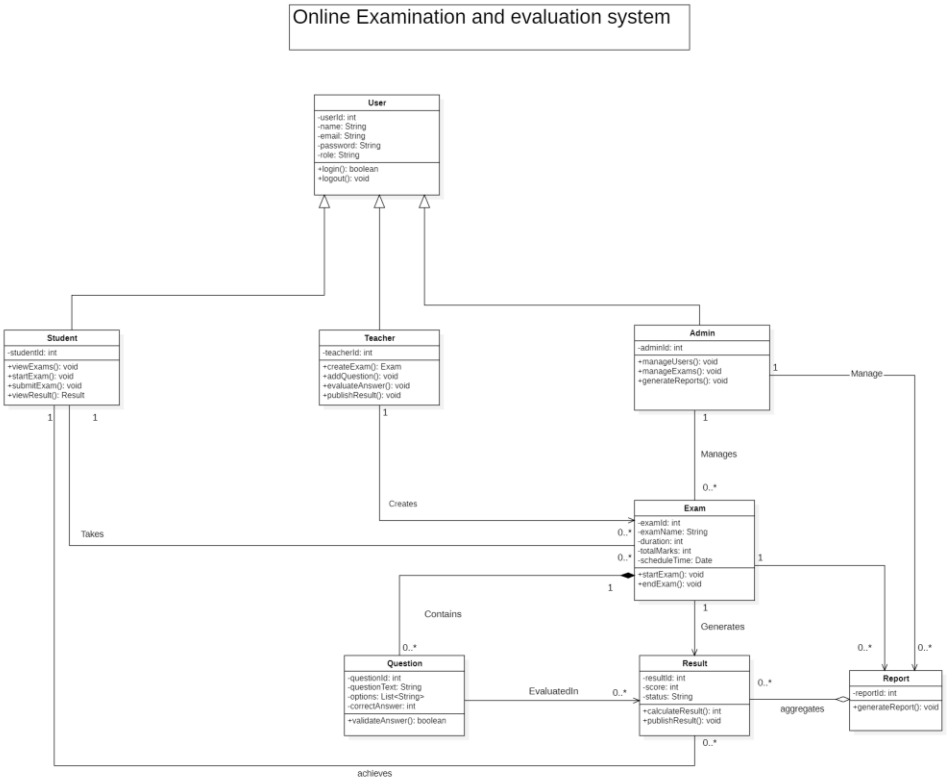


3.Creating Exam:



4.User Management:



B.7 Class Diagram

The Class Diagram represents the static structure of the system by showing classes, attributes, and relationships.